MUHAMAD FAJAR

Data Science | Production Operations | Process Improvement

+62 823-1824-2346 • fajrm238@gmail.com • Github: fajar431 • Depok, Indonesia

PROFESSIONAL SUMMARY

Results-driven professional with over 9 years of experience in manufacturing operations and production management at PT. Xacti Indonesia. Currently transitioning into Data Science, with hands-on training in Python, Machine Learning, and data analysis through Purwadhika Digital Technology School. Adept at process optimization, data documentation, cross-functional coordination, and SOP compliance. Seeking to leverage both operational expertise and emerging data science skills to drive business insights and operational efficiency.

SKILLS

Data Science & Analytics: Python, Pandas, NumPy, Scikit-learn, Data Visualization (Matplotlib, Seaborn), SQL, Machine Learning, EDA, Cohort Analysis

Operations & Production: Production Planning, SOP Development & Compliance, Quality Control, Inventory Management (FIFO), Scheduling, Reporting

Tools & Software: Microsoft Excel, Microsoft Word, Google Sheets, ERP Systems

Soft Skills: Attention to Detail, Cross-functional Coordination, Problem Solving, Time Management, Analytical Thinking

Language: Indonesian (Native), English (Professional Working Proficiency)

PROFESSIONAL EXPERIENCE

Production Staff | **PT. Xacti Indonesia** *2020 – 2025*

- Prepared, managed, and maintained production-related documents including work orders, schedules, and operational reports.
- Accurately entered and updated production data into the company ERP system on a timely basis, ensuring data integrity.
- Monitored and tracked production progress to ensure schedules and targets were consistently met.
- Coordinated with production supervisors and cross-functional departments regarding material availability, production planning, and delivery schedules.
- Prepared daily, weekly, and monthly production reports to support management decision-making.
- Supported the implementation of Standard Operating Procedures (SOPs) and ensured compliance with company policies.

Production Operator | **PT. Xacti Indonesia** *2016 – 2020*

- Operated and monitored production machinery and equipment in accordance with company standards and safety protocols.
- Ensured all products met quality and safety requirements before advancing to subsequent production stages.
- Followed SOPs and production schedules to consistently achieve daily output targets.
- Reported and documented issues, defects, or deviations identified during the production process.
- Maintained strict compliance with health, safety, and environmental regulations at all times.

PROJECTS & CAPSTONE

Hotel Booking Cancellation Prediction *2025 – 2026*

Stack: *Python, Pandas, Scikit-learn, Classification Modeling, Feature Engineering*

- Built a machine learning classification model to predict hotel booking cancellations using a historical dataset of 83,573 reservations from a hotel in Portugal.
- Achieved an F2-Score of 0.7665, successfully detecting 78.5% of cancellations with high recall optimized for business impact.
- Delivered a projected business impact of +$991,740 in potentially recovered revenue from 16,715 bookings in the test set alone.
- Applied feature engineering, class imbalance handling, and model evaluation techniques to optimize predictive performance.

Cohort Analysis – Customer Retention & Revenue Performance (TheLook E-Commerce) *2025 – 2026*

Stack: *Python, Pandas, SQL, Matplotlib, Seaborn, Cohort Analysis*

- Analyzed customer retention and revenue retention patterns across cohorts for TheLook e-commerce platform over the 2019–2024 period.
- Identified high-performing cohorts with the most sustainable retention and revenue contribution to support business growth strategy.
- Produced actionable data-driven recommendations for customer lifecycle management and retention campaigns.

Transjakarta Route Optimization – Operational Data Analysis *2025 – 2026*

Stack: *Python, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis*

- Conducted end-to-end data analysis on Transjakarta transit data to evaluate route performance and passenger behavior patterns.
- Generated data-driven operational optimization recommendations aimed at improving route efficiency and passenger experience.

Production Inventory Control System (CRUD + FIFO) *2025 – 2026*

Stack: *Python, Data Structures, FIFO Algorithm, CRUD Operations*

- Developed a Python-based inventory management system using FIFO (First-In, First-Out) data structure for component stock management.
- Implemented a warehouse approval workflow where production requests must be authorized before stock is released, simulating real-world production control.
- Demonstrated applied understanding of data structures, inventory logic, and production operations from professional manufacturing background.

M-Banking Mini – Terminal-Based Banking Simulation *2025 – 2026*

Stack: *Python, OOP, CLI Application*

- Built a terminal-based banking simulation application in Python featuring account registration, financial transactions, and inter-account transfers.
- Implemented core banking logic including balance management, transaction history, and user account administration using object-oriented programming.

EDUCATION & TRAINING

Purwadhika Digital Technology School *2025 – 2026*

Data Science & Machine Learning Bootcamp (Professional Training)

SMK Insan Kreatif *2008 – 2011*

Industrial Electronics (Elektronika Industri)

CERTIFICATIONS

- Data Science & Machine Learning Bootcamp — Purwadhika Digital Technology School (2025–2026)
- Industrial Electronics Technician — SMK Insan Kreatif (2008–2011)